

每日 MIR 論文精選 - 2026-05-23

優先序：MeanAudio / text-to-audio generation → karaoke-jp → genre/tagging ◦ Generated at 2026-05-23 12:55:29 Asia/Taipei.

#1 · MeanAudio / text-to-audio 主線

RobustSpeechFlow: Learning Robust Text-to-Speech Trajectories via Augmentation-based Contrastive Flow Matching

作者	Jinhyeok Yang, Hyeongju Kim, Yechan Yu, Joon Byun, Frederik Bous, Juheon Lee
日期	2026-05-21
來源	https://arxiv.org/abs/2605.22083v1
Topic tags	text-to-audio/music, flow/diffusion
品質判斷	新 · 新穎性 2 天內更新 · 關聯分數 mean=3 / kara=0

為什麼選它

以主線優先序、近期性、關鍵詞相關性與去重紀錄做初篩；這篇在今天的候選池中相對值得快速掃過。

摘要

While flow-matching text-to-speech (TTS) achieves strong zero-shot speaker similarity and naturalness, it remains susceptible to content fidelity issues, particularly skip and repeat errors from imperfect alignment. We propose RobustSpeechFlow, a training strategy that improves alignment robustness by extending contrastive flow matching with length-preserving repeat and skip latent augmentations. Requiring no external aligners or preference data, our method directly penalizes realistic failure modes and readily integrates into existing pipelines. On Seed-TTS-eval, it reduces the word error rate (WER) from 1.44 to 1.38 using only 0.06B parameters. On our ZERO500 benchmark, it delivers consistent intelligibility improvements across diverse speaker and prosody conditions; at NFE=24, it reduces English

character error rate (CER) from 0.48\% to 0.35\% and Korean CER from 0.81\% to 0.57\%.
Audio samples: <https://robustspeechflow.github.io/>

主要貢獻（快速驗證清單）

先看問題定義、baseline、評估資料、音訊樣本或程式碼是否可信，再決定是否深讀。

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優先檢查它能否影響 MeanAudio / text-to-audio 生成的資料與 caption 策略、訓練穩定性、評估方法、速度/RTF、codec 或 reward alignment。

對研究/實作的啟發

優先抽出可做成小實驗的部分：資料處理、caption 策略、評估指標、速度/可控性取捨、或對齊方法。

限制與注意事項

如果只依賴單一自動化指標，尤其 text-audio similarity 類，請用實際樣本、ablation 與多指標交叉檢查。

#2 · MeanAudio / text-to-audio 主線

Instrumental Text-to-Music Generation with Auxiliary Conditioning Branches

作者	Junyoung Koh
日期	2026-05-20
來源	https://arxiv.org/abs/2605.21433v1
Topic tags	text-to-audio/music
品質判斷	新 · 新穎性 3 天內更新 · 關聯分數 mean=3 / kara=0

為什麼選它

以主線優先序、近期性、關鍵詞相關性與去重紀錄做初篩；這篇在今天的候選池中相對值得快速掃過。

摘要

Text-to-music generation has advanced rapidly, with modern autoregressive and diffusion-based models producing convincing music from natural-language prompts. However, much of this progress relies on large-scale training data and external pretraining, making it difficult to isolate which design choices remain effective when data and pretraining are controlled. We study this setting using a Diffusion Transformer backbone with lyric and timbre conditioning, adapted to an instrumental-only text-to-music task in which the auxiliary lyric and timbre branches receive only degenerate conditioning signals. Through controlled ablations, we find that models retrained without these branches score lower across AudioBox aesthetics, LLM-as-judge, and human MOS, and that reinvesting the saved parameters as additional DiT depth recovers only marginally. This suggests the auxiliary branches may act as training-time architectural anchors whose contribution goes beyond their explicit conditioning content. We validate the same model through comparisons with external instrumental baselines and through our submission to the ICME 2026 Academic Text-to-Music (ATTM) Grand Challenge, where our Performance submission ranked first under both the objective metrics and the subsequent organizer-administered MOS over 35 raters, attaining the highest overall MOS across all challenge submissions, while our Efficiency submission was a finalist that tied for second under the objective metrics.

主要貢獻（快速驗證清單）

先看問題定義、baseline、評估資料、音訊樣本或程式碼是否可信，再決定是否深讀。

和你的研究/專案的關係

優先檢查它能否影響 MeanAudio / text-to-audio 生成的資料與 caption 策略、訓練穩定性、評估方法、速度/RTF、codec 或 reward alignment。

對研究/實作的啟發

優先抽出可做成小實驗的部分：資料處理、caption 策略、評估指標、速度/可控性取舍、或對齊方法。

限制與注意事項

如果只依賴單一自動化指標，尤其 text-audio similarity 類，請用實際樣本、ablation 與多指標交叉檢查。

今天的 take-away

先用「能不能轉成你下一個小實驗」來讀：資料/caption、評估、速度/RTF、可控性與對齊。如果論文依賴相似度類指標，請補實際聆聽、多指標與 ablation，避免被漂亮分數帶偏。

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